METHODOLOGY REPORT

ARC Congestion and Bottleneck Identification (CBI)

Comprehensive Technical Methodology

Analytical Framework, Statistical Diagnostics, and Validation

---

Organization

Atlanta Regional Commission

Date

October 2026

Technical Report on Arterial and Freeway Congestion Analytics

Contents

Executive summary **3**

1 Data architecture and analytical inputs **3**
1.1 Separation of observed and estimated information 4

2 Roadway network preparation and TMC eligibility **4**
2.1 Network metadata recovery 4
2.2 Road-name normalization 4
2.3 Spatial and directional integrity 4

3 Replica directional AADT and traffic-volume estimation **5**
3.1 Replica-to-TMC conflation 5
3.2 NPMRDS fallback 5
3.3 Traffic-volume uncertainty 5

4 GDOT hourly distribution factors **6**
4.1 Source and construction 6
4.2 Mathematical definition 6
4.3 Application to Replica directional AADT 6
4.4 Important assumptions 7

5 Free-flow speed estimation and excess travel time **7**
5.1 Excess travel time per vehicle 7
5.2 Excess vehicle-hours of delay 7
5.3 Annual interpretation 8

6 Corridor construction, Area Type, and spatial aggregation **8**
6.1 Area Type classification 8
6.2 Integration into CBI 8
6.3 Section construction algorithm 9
6.4 Example 9
6.5 Role in congestion analysis 9

7 Corridor severity, EAVHD density, and localized hotspots **10**
7.1 Corridor EAVHD 10
7.2 EAVHD density 10
7.3 Localized hotspot analysis 10

8 Recurring congestion and bottleneck identification **10**
8.1 Speed ratio 11
8.2 Freeway time–space event detection 11
8.3 Recurring bottleneck locations 11
8.4 Important qualification 12

9 Congestion Operating Profiles: K-means clustering **12**
9.1 Feature construction 12
9.2 K-means objective 12
9.3 Production configuration 12

10 Silhouette analysis and alternative clustering **13**
10.1 Temporal smoothing experiment . . . 13

11 Statistical analysis of Area Type and congestion severity **14**
11.1 Spearman rank correlation . . . 14
11.2 Kendall rank correlation . . . 14
11.3 Kruskal–Wallis testing . . . 14
11.4 Holm-adjusted pairwise testing . . . 14
11.5 Partial correlations and robust OLS . . . 14

12 Freeway–arterial coupling and lagged congestion relationships **15**
12.1 Candidate pairing . . . 15
12.2 Temporal lags . . . 15
12.3 Phi coefficient . . . 15
12.4 Fisher's exact test . . . 15
12.5 False Discovery Rate correction . . . 15
12.6 Illustrative findings from the recorded analysis . . . 16

13 Nine-component Speed Explorer **16**
13.1 Why multiple views matter . . . 17
13.2 Visual diagnostics versus empirical validation . . . 17

14 Calibration, verification, and validation **18**
14.1 Calibration . . . 18
14.2 Computational verification . . . 18
14.3 Internal diagnostic validation . . . 18
14.4 Independent empirical validation . . . 19

15 Recommended formal validation matrix **19**
15.1 Ranking stability . . . 20

16 How the CBI components work together **20**

17 Relevance to transportation planning and federal performance-based decision-making **21**
17.1 Reproducibility and transparency . . . 21

18 Overall methodological assessment **21**

19 Recommended conclusion for the technical report **22**

Source documentation **22**

Executive summary

The Congestion and Bottleneck Identification (CBI) project is an integrated transportation analytics framework developed to identify, quantify, classify, and visualize recurring congestion across the Atlanta Regional Commission's (ARC) 21-county metropolitan planning region.

The framework combines observed roadway operating conditions, estimated directional traffic volumes, empirically derived hourly traffic distributions, roadway geometry, and spatial classifications from ARC's Activity-Based Model (ABM). It extends conventional congestion reporting through temporal analysis, statistical inference, unsupervised machine learning, spatial bottleneck identification, and interactive visualization.

The project is organized around several analytical objectives:

1. Identify where congestion occurs, how frequently it occurs, and how long it persists.
2. Quantify the magnitude of congestion using speed deterioration, excess travel time, and estimated excess vehicle-hours of delay.
3. Distinguish localized recurring bottlenecks from broader corridor-level congestion.
4. Classify arterial operating conditions according to recurring temporal patterns.
5. Examine statistical relationships between freeway congestion and nearby arterial operations.
6. Incorporate land-use and development context from ARC's ABM into arterial segmentation and interpretation.
7. Support transparent and reproducible quality assurance through statistical diagnostics and the nine-component Speed Explorer.

The principal methodological advantage is the separation of **observed traffic operating conditions**, **estimated traffic exposure**, **spatial context**, and **analytical interpretation**. These components are integrated into a common system without treating estimated traffic volumes as direct observations or treating visualization as a substitute for empirical validation.

The following sections distinguish implemented procedures, documented parameters, exploratory analyses, and additional validation that remains necessary.

1. Data architecture and analytical inputs

The CBI framework integrates multiple transportation datasets. Each provides a different analytical function.

Data source	Principal contribution
INRIX / NPMRDS	Five-minute probe speeds, TMC identifiers, roadway lengths, reference speeds, and network metadata
Replica 2025	Estimated directional AADT associated with TMC roadway segments
GDOT 2025 TMAS	Observed hourly traffic counts used to derive hourly distribution factors
ARC ABM	Area Type classification describing land-use and development intensity

Data source	Principal contribution
TMC geometry	Spatial alignment, directionality, segment connectivity, and corridor construction
NPMRDS/HPMS metadata	Supplemental traffic-volume estimates and roadway classification
PostgreSQL / PostGIS	Storage, relational processing, spatial analysis, and analytical integration

1.1 Separation of observed and estimated information

This distinction is essential to the credibility of the project.

Observed inputs include GDOT traffic-count measurements and INRIX/NPMRDS probe-speed observations. Probe speeds are measurements derived from sampled vehicles, not necessarily the speeds of every vehicle using a road.

Estimated inputs include Replica directional AADT, modeled hourly traffic volumes, and some fallback traffic-volume estimates.

Derived analytical measures include excess travel time, EAVHD, recurrence, congestion severity, cluster assignments, and corridor rankings.

This separation allows analysts to identify which results depend on direct observations and which depend on modeling assumptions.

2. Roadway network preparation and TMC eligibility

The TMC is the fundamental spatial observation unit in the CBI framework. Each TMC represents a directional roadway segment associated with speed observations and network metadata.

For arterial analysis, the documented implementation selects principal and minor arterials using the applicable roadway functional-system codes, including the code's `f_system` values of 3 and 4.

2.1 Network metadata recovery

Some TMC records have incomplete functional classification or traffic metadata. The implementation includes a metadata backfill procedure using related TMC-linear records. The process applies safeguards to prevent limited-access freeway classifications from being incorrectly assigned to surface-street arterials.

2.2 Road-name normalization

Road names may differ across counties and datasets. A state route may be identified by its route number in one location and by its local street name elsewhere. The framework therefore includes road-name normalization and curated aliases to improve consistency in corridor construction.

2.3 Spatial and directional integrity

TMCs are associated with roadway geometry, direction, and spatial ordering. These attributes support:

- Correctly ordered corridor segments.
- Separate examination of opposing travel directions.
- Detection of disconnected roadway sections.
- Identification of localized bottlenecks.
- Aggregation of individual TMC measurements into meaningful analytical sections.

The objective is to avoid treating a long roadway as one homogeneous congestion location when its operational characteristics vary substantially.

3. Replica directional AADT and traffic-volume estimation

AADT provides the traffic-exposure component of the CBI delay calculations. The arterial congestion implementation uses Replica 2025 directional AADT as the primary volume source where suitable matches are available.

The documented default configuration is:

```
AADT_SOURCE = "replica"
REPLICA_DIR_AADT_FROM = "table"
```

The principal database source is `Year_2025.replica_tmc_dir_aadt`.

3.1 Replica-to-TMC conflation

Replica traffic-volume estimates are spatially associated with directional TMCs. This requires reconciling differences between the Replica roadway network and the INRIX/NPMRDS TMC network. The conflation process must account for directional compatibility, roadway class, geometry, and possible differences in segment length.

Not every TMC necessarily receives a reliable Replica match. The pipeline therefore includes fallback and quality-screening procedures.

3.2 NPMRDS fallback

Where a reliable Replica directional estimate is unavailable, the methodology may use NPMRDS traffic-volume metadata. The important distinction is that a two-way AADT estimate requires a directional allocation, while a directional Replica AADT does not.

$$AADT_{i,dir} = AADT_{i,2way} \times D_i, \qquad D_i = 0.50.$$

This is an assumed equal directional split, not an empirically measured directional distribution.

3.3 Traffic-volume uncertainty

Because EAVHD is proportional to estimated vehicle volume when all other inputs are fixed, errors in directional AADT propagate directly into delay estimates. For example, a 20% overestimate of directional AADT produces a 20% overestimate of the associated volume-weighted delay, assuming the speed and travel-time inputs remain unchanged.

Consequently, Replica matching quality, fallback provenance, and traffic-volume uncertainty are important elements of the CBI validation framework.

4. GDOT hourly distribution factors

A major component of the arterial methodology is the development of regionally relevant hourly distribution factors from GDOT traffic-count observations.

AADT represents an average daily traffic volume. It does not, by itself, indicate how traffic is distributed across the 24 hours of a day. The Hourly Distribution Factor (HDF) bridges this gap.

4.1 Source and construction

The documented HDF development uses 2025 Georgia TMAS continuous traffic-count data from 23 urban arterial stations within the ARC region.

The source records include principal and minor arterial monitoring stations, with invalid sensor days screened before constructing representative hourly distributions.

The reported procedure retained approximately 99.7% of station-days and used station/day-of-week weighting to avoid allowing stations with more records to dominate the profile.

These are code-documented results; the original station-level audit should be retained for independent verification.

4.2 Mathematical definition

For station s, day d, and hour h:

$$HDF_{s,d,h} = \frac{V_{s,d,h}}{\sum_{k=0}^{23} V_{s,d,k}}$$

where $V_{s,d,h}$ represents observed hourly traffic. The resulting station-day profiles are aggregated into a representative 24-hour distribution. The final profile is normalized so that:

$$\sum_{h=0}^{23} HDF_h = 1$$

when expressed as fractions.

4.3 Application to Replica directional AADT

For a TMC with Replica directional AADT:

$$\boxed{\hat{V}_{i,h} = AADT_{i,dir}^{Replica} \times HDF_h}$$

For an illustrative directional AADT of 40,000 vehicles/day and an HDF of 7.5%:

$$\hat{V}_{i,17} = 40,000 \times 0.075$$

$$\boxed{\hat{V}_{i,17} = 3,000 \text{ vehicles/hour}}$$

Because Replica AADT is already directional, no additional 0.50 adjustment should be applied.

For a two-way NPMRDS fallback:

$$\hat{V}_{i,h} = AADT_{i,2way}^{NPMRDS} \times 0.50 \times HDF_h.$$

4.4 Important assumptions

The resulting hourly volumes are estimated rather than directly observed on each TMC.

The procedure assumes that the representative GDOT hourly distribution is reasonably transferable to the TMCs receiving it.

It also assumes that the selected traffic-volume source and hourly profile are compatible in temporal definition, directional interpretation, and roadway coverage.

These assumptions should be evaluated through sensitivity analysis and comparisons with independent hourly counts.

5. Free-flow speed estimation and excess travel time

The arterial CBI uses a segment-specific free-flow reference to measure deterioration in operating speed.

The documented arterial implementation identifies a common reference hour based on the highest regional mean hourly speed. Each TMC then receives its own reference speed corresponding to that hour.

A safeguard examines cases where the resulting baseline is implausibly low and can substitute a higher reference based on the segment's 85th-percentile hourly-average speed.

This addresses situations in which an anomalous overnight speed would otherwise make daytime congestion appear artificially mild.

5.1 Excess travel time per vehicle

For TMC i at time t:

$$ETT_{i,t} = L_i \max\left(0, \frac{1}{S_{obs,i,t}} - \frac{1}{S_{FF,i}}\right)$$

where:

- L_i: TMC length in miles.
- $S_{obs,i,t}$: observed speed in miles per hour.
- $S_{FF,i}$: reference free-flow speed in miles per hour.
- $ETT_{i,t}$: excess travel time in hours per vehicle.

The maximum operator prevents speeds above the reference from generating negative delay.

5.2 Excess vehicle-hours of delay

The estimated excess vehicle-hours of delay is:

$$\boxed{EAVHD = \sum_i \sum_t \hat{V}_{i,t} ETT_{i,t}}$$

For a five-minute time interval, if hourly traffic is assumed to be uniformly distributed within that hour:

$$\hat{V}_{i,t}^{5min} = \frac{\hat{V}_{i,h}}{12}.$$

This provides a consistent relationship between five-minute speed observations and estimated traffic exposure.

For hourly aggregation, the corresponding hourly estimate is the sum of the interval-level volume-weighted excess travel times.

5.3 Annual interpretation

An annual EAVHD estimate requires appropriate accounting for the number of observed days, missing observations, and any annualization procedure.

An observed-period delay total should not be labeled an annual total unless its temporal coverage and expansion method support that interpretation.

The documented implementation includes a minimum analyzed-days threshold of 30 for relevant arterial metrics. That is a data-support rule, not proof of full-year representativeness.

6. Corridor construction, Area Type, and spatial aggregation

One of the important additions to this arterial analytics is the integration of Area Type from ARC's ABM.

The purpose is to prevent analytical corridors from combining roadway environments that differ substantially in development intensity and operating context.

6.1 Area Type classification

The ABM-derived `ATYPE` attribute identifies seven development contexts.

Code	Area Type
1	Very high-density urban / CBD
2	High-density urban
3	Medium-density urban
4	Low-density urban
5	Suburban
6	Exurban
7	Rural

These categories describe the built environment, not congestion severity.

The upstream ABM classification is associated with land-use and development characteristics, including population and employment context.

The exact PCS input fields and numerical thresholds used to generate each class have not been independently established from the methodology records examined here. They should be verified against the ABM's authoritative area-type generation procedure before being published as exact definitions.

6.2 Integration into CBI

The documented data lineage carries ABM-derived Area Type through `TMC_AreaType.geojson` and the corresponding roadway/TMC attribution into the arterial analytical data.

The production configuration includes:

```
SPLIT_BY_AREA_TYPE = True
SPLIT_BY_STREET_NAME = True
MIN_SECTION_MILES = 1.0
MAX_PAIRING_METRES = 800
```

6.3 Section construction algorithm

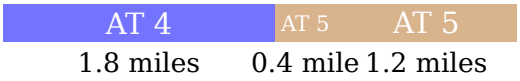
For each road:

1. Identify the reference direction, based on the direction with more analyzed miles.
2. Order TMCs spatially along that direction.
3. Identify transitions in Area Type and local street name.
4. Construct preliminary contiguous sections.
5. Merge sections shorter than one mile using the documented neighboring-run rule.
6. Associate opposite-direction TMCs with nearby reference sections, subject to the 800-meter pairing safeguard.
7. Assign the resulting section its mileage-dominant Area Type while retaining its Area Type composition.

Each TMC remains intact as an analytical observation. A TMC crossing an Area Type boundary is not fractionally divided for speed calculations.

6.4 Example

Consider a hypothetical arterial with these contiguous runs:



The 0.4-mile run is shorter than the minimum. Under the documented rule, it merges toward the longer neighboring run, even if that neighbor has a different Area Type.

The resulting section may contain more than one Area Type. Therefore, the dominant classification should not be interpreted as indicating that every TMC has the same development context.

6.5 Role in congestion analysis

Area Type directly affects section boundaries and the spatial units used for aggregation.

It does not directly modify:

- Replica AADT.
- GDOT hourly distribution factors.
- Free-flow speed.
- TMC-level excess travel time.
- The mathematical EAVHD formula.

Its effect on corridor rankings is indirect, through section membership, length, and aggregation.

This is best described as **ABM-informed geographic stratification**, not empirical calibration of the traffic-volume or delay model.

7. Corridor severity, EAVHD density, and localized hotspots

A long arterial can accumulate substantial delay simply because it contains many miles of roadway.

This framework addresses this by distinguishing total estimated delay from normalized delay density.

7.1 Corridor EAVHD

For a section c:

$$EAVHD_c = \sum_{i \in c} EAVHD_i.$$

This represents the estimated aggregate delay across the section's TMCs.

7.2 EAVHD density

The documented ranking uses:

$$\boxed{Density_c = \frac{EAVHD_c}{L_{c,unique}}}.$$

The denominator represents unique analyzed corridor mileage, based on the longest direction, rather than simply adding the mileage of both opposing directions.

This helps reduce the tendency for long corridors to dominate rankings purely because of their length.

The arterial ranking eligibility rules include a minimum of one unique mile and at least four non-stub TMCs, with a 0.05-mile stub threshold.

7.3 Localized hotspot analysis

The framework also uses a rolling one-mile hotspot window.

This is important because an arterial with relatively modest average corridor-wide delay may contain a short, highly congested location.

The localized hotspot measure can identify that concentration without allowing the remaining uncongested miles to dilute its severity.

A half-mile alternative is retained for sensitivity testing.

The corridor-wide and localized hotspot measures answer different questions:

- Corridor density: How severe is congestion across the overall analytical section?
- Localized hotspot: Where is the most concentrated operational problem within the section?

These measures should be reported separately rather than treated as interchangeable rankings.

8. Recurring congestion and bottleneck identification

The freeway and arterial pipelines share speed-based congestion concepts but use different detection procedures.

8.1 Speed ratio

A fundamental congestion indicator is:

$$R_{i,t} = \frac{S_{obs,i,t}}{S_{ref,i}}.$$

A commonly used screening condition in this implementation is:

$$R_{i,t} < 0.70.$$

This indicates observed speeds below 70% of the selected reference speed.

The 70% threshold is a congestion-screening rule. It is not the trigger used in the continuous EAVHD calculation, which can register positive delay at less severe speed deterioration.

8.2 Freeway time–space event detection

This freeway event detector organizes five-minute congestion observations into a time-by-segment matrix.

Connected congestion cells are grouped into events using eight-connected component labeling, allowing continuity across adjacent roadway segments and neighboring time intervals.

The documented screening rules include a minimum of six event cells, at least ten minutes of duration, at least two segments, and minimum spatial support of 0.25 mile.

A five-mile maximum active extent safeguard limits implausibly merged congestion events.

These parameters help exclude isolated observations and processing artifacts. They should be characterized as implemented screening thresholds unless independently calibrated against verified bottleneck events.

8.3 Recurring bottleneck locations

For recurring bottlenecks, the framework calculates spatial recurrence profiles and applies smoothing before detecting significant local peaks.

The documented freeway settings include:

Parameter	Implemented value
Spatial smoothing window	3 TMC segments
Minimum peak occurrence	20%
Minimum peak prominence	10 percentage points
Minimum peak separation	3 segments
Boundary threshold	50% of local peak recurrence

These rules identify distinct recurring congestion concentrations along a corridor.

A roadway may therefore contain multiple bottlenecks, each with its own frequency, duration, location, and spatial extent.

8.4 Important qualification

The three-segment smoothing window does not represent a constant physical distance because TMC lengths differ.

Also, the documented recurring-peak constants were reconstructed from earlier report behavior and require reconciliation with the original validated implementation before being described as empirically calibrated.

9. Congestion Operating Profiles: K-means clustering

This arterial CBI includes an unsupervised machine-learning component to classify recurring daily congestion patterns.

This is distinct from ranking roadways by total delay.

The clustering process groups TMCs based on similarities in their normalized hourly speed profiles.

9.1 Feature construction

For each TMC, observed speeds are normalized relative to the segment-specific free-flow baseline:

$$R_{i,h} = \frac{S_{i,h}}{S_{FF,i}}.$$

The documented COP implementation uses daytime features from 06:00 to 20:00, with AM and PM summary periods.

Residual speed ratios are capped at 1.2 for clustering to limit the influence of unusually high ratios.

This cap applies to COP features, not the core EAVHD computation.

9.2 K-means objective

K-means partitions TMC observations into K clusters by minimizing within-cluster squared distances:

$$\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2.$$

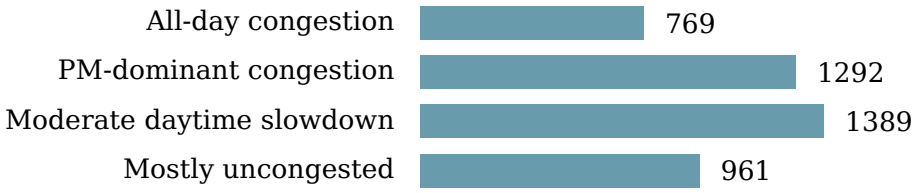
Here, x_i represents a TMC's congestion profile and μ_k is its cluster centroid.

9.3 Production configuration

The October 2026 comprehensive methodology records a production comparison selecting K-means with four clusters and a silhouette score of approximately 0.293.

Clusters: 4	Silhouette: 0.293	TMC profiles: 4,411

The documented profile categories and counts are:



These clusters allow analysts to distinguish roads with sustained congestion from those with more concentrated peak-period slowdowns.

However, the recorded centroids have broadly similar daily shapes, meaning that the clusters may reflect differences in congestion intensity as well as differences in temporal pattern.

This limitation should be acknowledged when interpreting the COP labels.

10. Silhouette analysis and alternative clustering

Silhouette analysis evaluates the internal separation of the clusters.

For observation i:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where $a(i)$ represents average within-cluster distance and $b(i)$ represents the average distance to the nearest alternative cluster.

A score near 1 indicates strong separation, while a score near 0 indicates overlapping clusters.

The recorded score of 0.293 indicates modest separation rather than sharply differentiated groups.

The October methodology records a hierarchical clustering comparison at $K = 4$, with a silhouette score of approximately 0.256.

That comparison supported the selection of K-means for the recorded production configuration.

Earlier experimental scores should not be mixed with the production comparison without identifying the exact dataset, feature transformations, and run configuration.

10.1 Temporal smoothing experiment

We also developed an experimental three-point centered moving average:

$$R_{smooth}(h) = \frac{R(h-1) + R(h) + R(h+1)}{3}$$

where all three observations are available.

This reduces short-duration fluctuations in the daily speed profile.

The experimental procedure compares raw and smoothed profiles using silhouette scores, cluster assignments, transition matrices, and centroid shapes.

Smoothing can improve the apparent stability of clusters, but it can also suppress meaningful short-duration congestion peaks.

For that reason, the smoothing experiment is diagnostic and should not be described as active production processing unless confirmed by the corresponding production run.

Importantly, the documented procedure does not apply this COP smoothing to EAVHD.

11. Statistical analysis of Area Type and congestion severity

In addition to using ABM Area Type for geographic sectioning, we developed exploratory statistical tests examining its association with congestion characteristics.

The diagnostic module is `cbi_area_type_severity_tests.py`.

11.1 Spearman rank correlation

Spearman correlation evaluates monotonic associations between ranked variables.

It can help determine whether congestion severity tends to increase or decrease across an ordered representation of development intensity.

However, the numerical Area Type codes must be interpreted carefully because the difference between adjacent codes does not necessarily represent an equal change in density.

11.2 Kendall rank correlation

Kendall's tau provides another rank-based measure of association.

It can be useful where there are tied observations or where a complementary nonparametric assessment is desirable.

11.3 Kruskal–Wallis testing

The Kruskal–Wallis test examines whether congestion severity distributions differ across multiple groups.

For example, it can test whether the distribution of EAVHD density differs among urban, suburban, exurban, and rural groups.

A significant result indicates differences among groups but does not establish which specific groups differ.

11.4 Holm-adjusted pairwise testing

Pairwise comparisons can investigate which groups exhibit statistically distinguishable severity distributions.

Holm adjustment controls the family-wise error rate across those multiple comparisons.

11.5 Partial correlations and robust OLS

Partial rank correlations can examine associations while accounting for selected additional variables.

Robust ordinary least-squares models can evaluate conditional relationships between congestion measures and explanatory variables, with appropriate handling of heteroskedasticity or outliers depending on the estimator.

These procedures add analytical depth, but the documented script is exploratory and is not part of the production EAVHD estimation process.

No specific significant Area Type effect should be claimed without the retained test outputs, model specifications, and diagnostic results.

12. Freeway–arterial coupling and lagged congestion relationships

This project also investigates whether congestion on arterials is associated with congestion on nearby freeways.

This addresses an important regional network question: whether congestion patterns are spatially and temporally related across roadway facilities.

12.1 Candidate pairing

Arterial hotspots are paired with nearby freeway corridors using geographic proximity or roadway/intersection name matching.

These procedures identify candidate relationships. They do not by themselves establish a direct traffic connection.

12.2 Temporal lags

The documented analysis evaluates freeway-leading lags of zero hours, one hour, and two hours.

This allows investigation of simultaneous and delayed congestion associations.

12.3 Phi coefficient

For binary congestion indicators, the phi coefficient measures association strength.

$$\phi = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}$$

where a, b, c, d are the counts in a two-by-two contingency table.

Values near zero indicate weak binary association. Larger absolute values indicate stronger association.

12.4 Fisher's exact test

Fisher's exact test evaluates evidence against independence between the two binary congestion indicators.

It is useful for contingency tables, including those with relatively small cell counts.

12.5 False Discovery Rate correction

Because many hotspot/freeway combinations and time lags are tested, the analysis faces a multiple-comparisons problem.

We used the Benjamini–Hochberg False Discovery Rate procedure to adjust for multiple testing.

For ordered p-values $p_{(1)}, \ldots, p_{(m)}$, the procedure compares them against:

$$p_{(k)} \leq \frac{k}{m}\alpha$$

with a documented reporting threshold of:

$$q < 0.05.$$

FDR adjustment helps control the expected proportion of false discoveries among rejected hypotheses under the relevant assumptions.

12.6 Illustrative findings from the recorded analysis

The October methodology records a positive association between Pleasant Hill Road and I-85 southbound at a one-hour lag, with phi around 0.41–0.44.

It also records GA-138 and I-20 westbound with phi around 0.04, illustrating that a statistically significant adjusted result can still have negligible practical association.

These findings should be presented as exploratory statistical associations, not proof that freeway congestion caused arterial congestion.

Temporal autocorrelation, shared daily traffic cycles, and geographic pairing uncertainty remain important limitations.

The documented regional report configuration had the coupling display disabled pending further QA.

13. Nine-component Speed Explorer

The Speed Explorer is a major analytical and interpretive component of the CBI system.

It is not simply a collection of static maps. It provides multiple ways to interrogate the same underlying traffic observations and derived measures.

The documented system uses nine complementary views.

View	Description
1. Network	Displays spatial patterns of congestion and TMC coverage. It supports examination of roadway continuity, missing geometry, and hotspot locations.
2. Speed Classes	Displays normalized speed conditions using consistent speed-ratio thresholds, including 0.90, 0.70, 0.50, and 0.35.
3. Time of Day	Allows users to examine hourly conditions, animation, weekday/weekend contrasts, and time-dependent congestion patterns.
4. Delay & Volume	Combines estimated traffic exposure with excess travel time to show where substantial volume-weighted congestion occurs.
5. Direction	Examines differences between opposing travel directions and helps reveal potentially incorrect directional assignments.
6. Time–Space	Displays congestion as a function of location and hour, supporting examination of spatial discontinuities and temporal congestion propagation.
7. Segment Detail	Provides TMC-specific speed profiles, reference speeds, hourly metrics, and sample-support information.
8. 3D Delay Walls	Represents congestion severity through three-dimensional height and speed-ratio coloring, making concentrated delay patterns easier to identify visually.

View	Description
9. Hours Compared	Supports side-by-side examination of congestion conditions across hours using consistent visual scales and comparative heatmaps.

The prior chat included illustrative image-search references for these nine views, not screenshots of the actual Speed Explorer. This LaTeX edition preserves their descriptions without embedding unrelated third-party images.

13.1 Why multiple views matter

The nine views support complementary inspection of the same underlying analytical system.

For example, a severe congestion location identified in the Delay & Volume tab can be examined through:

- The Network tab to confirm its geographic location.
- The Direction tab to verify travel direction.
- The Time of Day tab to identify its operating period.
- The Time–Space tab to examine spatial and temporal continuity.
- Segment Detail to inspect the underlying TMC measurements.

This provides a structured analytical workflow for investigating unexpected results.

13.2 Visual diagnostics versus empirical validation

Visual inspection can reveal:

- Implausible speed ratios.
- Incorrectly connected TMCs.
- Directional mismatches.
- Unusual overnight congestion.
- Excessive or unexpected delay estimates.
- Missing observations.
- Spatial discontinuities.
- Inconsistent corridor aggregation.

However, visual consistency does not establish external accuracy.

A dashboard can reproduce an incorrect traffic-volume assumption consistently across all nine tabs.

Therefore, the Speed Explorer should be described as an **Integrated Visual Diagnostics and Analytical Quality Assurance Framework**.

It complements, but does not replace, independent empirical validation.

14. Calibration, verification, and validation

The CBI project includes multiple analytical safeguards, but these should be distinguished from completed empirical validation.

14.1 Calibration

Calibration involves selecting or adjusting model parameters using relevant empirical evidence.

Potential calibration subjects include:

- Free-flow speed baselines.
- Traffic-volume factors.
- Congestion screening thresholds.
- Spatial smoothing windows.
- Bottleneck detection parameters.
- Directional volume assumptions.

A parameter selected through engineering judgment is not necessarily empirically calibrated.

For example, the one-mile minimum arterial section length is an operational segmentation rule, not a statistically optimized parameter.

14.2 Computational verification

Verification examines whether the implemented calculations correctly follow the specified methodology.

Examples include:

- Confirming HDF values sum to 100%.
- Verifying that directional Replica AADT is not halved again.
- Recalculating EAVHD for selected TMCs.
- Checking units in travel-time calculations.
- Confirming correct aggregation of directional and spatial metrics.
- Ensuring Speed Explorer results match database outputs.

14.3 Internal diagnostic validation

This includes examining whether the model behaves plausibly.

Examples include:

- Testing anomalous free-flow baselines.
- Investigating unrealistic overnight congestion.
- Checking cluster stability.
- Examining changes in rankings under alternative HDF profiles.
- Evaluating sensitivity to section-length rules.
- Comparing different bottleneck detection thresholds.

14.4 Independent empirical validation

This is the strongest category.

It involves comparisons against information not used to construct or calibrate the relevant result.

Examples include:

- Replica-derived directional hourly volumes versus independent GDOT hourly counts.
- Identified bottleneck locations versus verified field observations.
- Modeled delay estimates versus independent travel-time and traffic-volume measurements.
- Freeway event detection versus independently labeled time–space congestion events.

The current methodology documentation supports substantial implemented diagnostics and safeguards, but does not establish that all these external validation exercises have been completed.

15. Recommended formal validation matrix

The following table distinguishes what the project has implemented from what should be retained or independently tested before making stronger accuracy claims.

Analytical component	Implemented basis	Additional validation evidence
TMC network	Functional-class screening, geometry, metadata backfill	Spatial completeness and classification audit
Replica AADT	Directional conflation and fallback rules	Independent directional count comparisons
GDOT HDF	23-station 2025 hourly profile	Station audit and held-out station comparisons
Free-flow baseline	Common reference hour and percentile safeguard	Independent speed-profile checks
EAVHD	Volume-weighted excess travel-time formula	Recomputed samples and external comparisons
Area Type	ABM-derived sectioning	Boundary, coverage, and rank-sensitivity audit
K-means	Four-cluster production configuration	Cluster stability and feature sensitivity
Silhouette	Internal cluster separation assessment	Repeated runs and alternative feature tests
FDR	Multiple-testing adjustment	Dependence-aware robustness testing
Bottleneck detection	Time–space events and spatial peaks	Independently labeled event comparison
Speed Explorer	Nine complementary analytical views	Documented visual QA findings and corrections

15.1 Ranking stability

Because CBI supports prioritization, one of the most important validation questions is whether rankings are stable under reasonable changes in analytical assumptions.

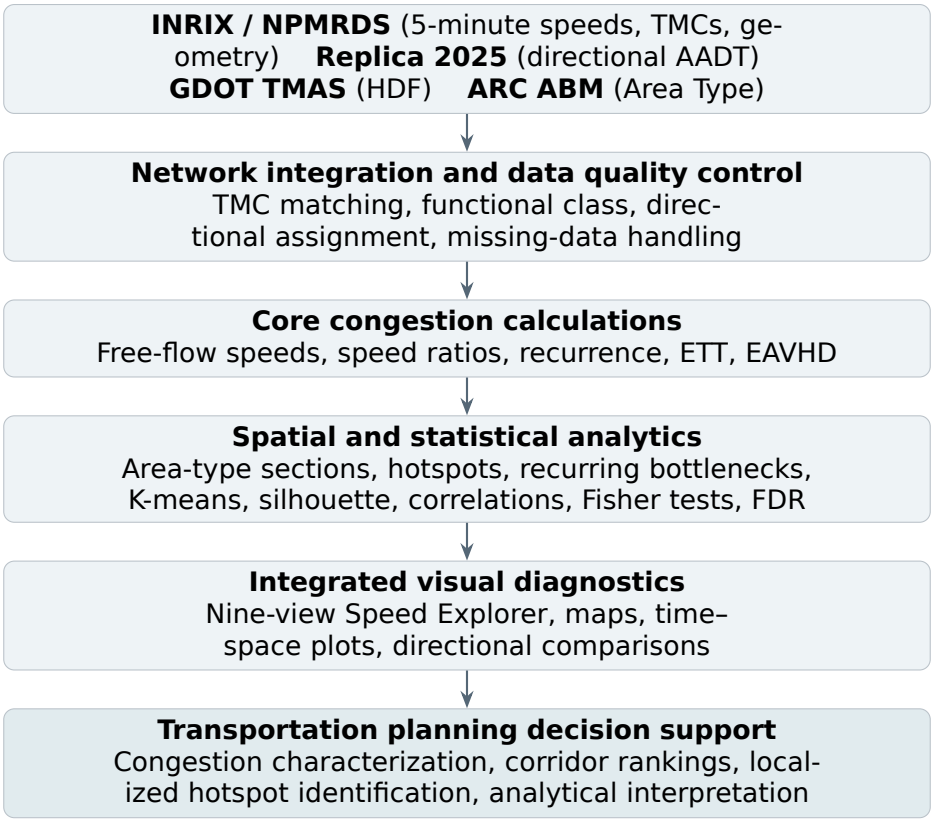
Recommended comparisons include:

- Alternative HDF profiles.
- Replica versus appropriate alternative AADT sources.
- Different free-flow reference assumptions.
- Alternative spatial smoothing windows.
- Different minimum section lengths.
- Alternative hotspot-window sizes.

For each scenario, calculate Spearman rank correlation, top-20 overlap, absolute rank changes, and changes in EAVHD density.

A strong ranking system should not change dramatically under small, reasonable adjustments unless the underlying data genuinely support those changes.

16. How the CBI components work together



17. Relevance to transportation planning and federal performance-based decision-making

The CBI framework provides a reproducible analytical basis for examining regional congestion patterns and identifying locations that may warrant further transportation planning or operational investigation.

Its principal planning contributions include more geographically specific bottleneck identification, time-dependent congestion characterization, traffic-exposure-weighted delay estimation, and transparent interpretation of regional roadway conditions.

These capabilities are relevant to performance-based metropolitan transportation planning under 23 CFR Part 450, including the evaluation of congestion conditions and potential transportation investments.

However, CBI outputs should not automatically be described as federally prescribed performance measures. Measures such as EAVHD density and COP membership may serve as supplemental planning indicators unless formally adopted for a particular performance-management purpose.

Similarly, the use of ABM Area Type does not mean that the entire CBI framework is itself an ABM forecasting application. It is an observational and estimated-exposure analytical system that incorporates an ABM-derived spatial classification.

17.1 Reproducibility and transparency

A defensible CBI evaluation should identify:

- The data vintage and sources.
- Model and script versions.
- Traffic-volume estimation procedures.
- Congestion thresholds.
- Spatial aggregation rules.
- Parameter choices.
- Missing-data treatment.
- Calibration and validation evidence.
- Known limitations and uncertainty.

These elements allow the methodology to be reviewed, repeated, and improved independently.

18. Overall methodological assessment

The CBI framework demonstrates substantial analytical development across transportation engineering, spatial data processing, statistical analysis, and machine learning.

Its strongest documented features are the integration of multiple traffic-data sources, directional TMC-based analysis, explicit delay formulas, ABM-informed section construction, recurring bottleneck detection, congestion operating-profile classification, and interactive diagnostic capabilities.

Its principal remaining methodological risks concern the independent accuracy of traffic-volume estimates, transferability of hourly factors, calibration of detection thresholds, spatial aggregation sensitivity, and external validation of congestion results.

The statistical components also require careful interpretation. K-means and silhouette analysis characterize internal data structure; FDR controls multiple-testing errors; and visualization supports diagnostics. None independently proves that the resulting traffic volumes, delay estimates, or project rankings are correct.

The appropriate next stage is therefore to consolidate the implemented methodology with a retained validation register containing actual production outputs, independent benchmarks, sensitivity tests, and documented QA findings.

19. Recommended conclusion for the technical report

Conclusion

The Congestion and Bottleneck Identification (CBI) framework developed for the Atlanta Regional Commission integrates observed probe-speed data, estimated directional traffic volumes, empirically derived hourly traffic distributions, and ABM-based spatial classifications into a comprehensive regional congestion analytics system.

The methodology employs TMC-level spatial and temporal processing, segment-specific free-flow speed estimation, excess vehicle-hours of delay calculations, area-type-informed corridor construction, localized hotspot analysis, and recurring bottleneck detection. These core procedures are supplemented by unsupervised machine learning, cluster-separation diagnostics, nonparametric statistical testing, multiple-comparison correction, and exploratory freeway–arterial congestion association analysis.

A nine-component Speed Explorer provides an integrated environment for spatial, temporal, directional, and segment-level examination of analytical results. The visualization framework supports data-quality assessment, anomaly identification, internal consistency checks, and detailed interpretation of congestion patterns.

Collectively, these capabilities establish a structured and reproducible methodology for identifying and characterizing congestion across the regional roadway network. The framework distinguishes observed operating conditions from modeled traffic exposure and preserves the distinction between analytical diagnostics, computational verification, and independent empirical validation.

Continued development should prioritize independent traffic-volume validation, documented parameter sensitivity, spatial and temporal coverage assessment, and quantitative verification of bottleneck identification and ranking stability. These activities will strengthen the framework’s reliability and defensibility as a transportation planning decision-support resource.

Source documentation

This write-up incorporates the project methodology discussed and the following existing CBI technical records:

- *CBI Comprehensive Technical Methodology and System Architecture*, October 2026.

- *CBI Detailed Methodology and Calibration Audit*, October 2026.
- *CBI Freeway Technical Methodology, Configuration, Calibration Evidence, and Validation Register*, October 2026.

Important documentation note: This is a comprehensive technical narrative grounded in previously documented scripts and methodology records, not a fresh execution of the PostgreSQL database or a completed independent validation study. Before formal publication, reconcile the production configuration and retain the actual output tables, logs, statistical results, and validation exhibits.